Can Large Language Models Extract Cross-Sectional Equity Signals from SEC Annual-Report Narratives?

A Validity-Driven Study Across Eleven Experimental Rounds

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Abstract

This report asks whether a large language model (LLM) reading the narrative sections of SEC 10-K filings can produce a signal that forecasts short-horizon relative equity returns. The study is organised as eleven experimental rounds. Each round exists to close one specific threat to the validity of the previous round's conclusion, rather than to try a new idea; more than half of the rounds changed how the result was *measured* rather than what was measured. The final design pairs a deterministic year-over-year text diff with an LLM that judges only whether a change is material, and evaluates the resulting signal through a sector-neutral rank-weighted long–short portfolio against three independent checks: a persistent neutralised random baseline, a Fama–French factor regression with Newey–West standard errors, and a sixfold expansion of the sample.

Three findings survive. First, at full statistical power the LLM "diff-surprise" signal on Item 1A Risk Factors is *significantly negative* (-0.83 Sharpe; FF3+MOM alpha -8.9%/yr, $t = -2.54$), worse than all 500 random baselines, and it reverses a positive result ($+0.57$ Sharpe, alpha $p = 0.045$) obtained on a smaller sample. Second, the failure is traceable to a structural negativity bias: the model used the full negative half of its permitted score range and never the top of the positive half. Third, moving the text source to the MD&A section did not rescue the approach—a pre-registered Lazy-Prices-style similarity hypothesis came out significantly *opposite* in sign (-0.72 Sharpe, alpha $t = -2.95$), a result that is robust within sample but is most plausibly survivorship bias in a current-membership universe.

The contribution is therefore not a profitable signal. It is a reusable validation framework, a documented record of the measurement errors it caught—including a parser defect affecting 42% of inputs and a filing-coverage defect that silently thinned the early cross-section to single digits—and a clean, well-powered negative answer to a question the literature leaves open.

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1 Introduction

1.1 Motivation

Public companies are legally required to describe, in narrative form, what could go wrong with their business (Item 1A, "Risk Factors") and how management reads its own results (Item 7, "Management's Discussion and Analysis", or MD&A). These sections are long, unstructured, and written by lawyers. They are also, in principle, the most explicitly forward-looking text a firm publishes on a fixed schedule.

Two literatures meet here. Accounting research has established that risk-factor disclosure is firm-specific and informative, and that *changes* in filing language from one year to the next carry return-relevant information. Separately, the recent LLM-in-finance literature has shown that language models can extract tradable sentiment from text. The obvious intersection—use an LLM to read the year-over-year change in a firm's risk narrative—has not, to our knowledge, been tested under proper cross-sectional controls.

The intersection is attractive precisely because it is cheap to reach: the data is free and public, the filing date gives an unambiguous point-in-time anchor, and the year-over-year structure means the model can be asked a narrow, well-posed question ("what is new, and does it matter?") rather than an open-ended one.

1.2 Research question

> **Can structured signals extracted by an LLM from SEC annual-report narratives forecast short-horizon relative stock returns, and do they beat a transparent lexicon baseline on identical data and identical portfolio construction?**

The comparison clause matters. An LLM signal that is merely positive is not interesting if a word-counting dictionary achieves the same thing on the same text; the question is whether semantic understanding buys anything over term frequency.

1.3 What this report is, and is not

This is a *research pipeline*, not a trading system. No transaction costs, borrowing costs, or capacity constraints are modelled, and no result here should be read as investment advice. The deliverable is a defensible answer and the apparatus that produced it.

A second framing point governs the whole report. In a study with no labelled ground truth and a per-call monetary cost, the binding difficulty is not getting the model to produce an answer—it is establishing whether any given answer should be believed. Accordingly, the organising unit of this report is the *threat to validity*: each round is presented as a specific threat that was identified, the method used to close it, and what closing it did to the conclusion.

1.4 Contributions

1. A point-in-time text-signal pipeline over 150 US large-cap names, 2011–2026, comprising 2,146 10-K filings with per-record input-quality instrumentation.

2. A hybrid extraction design that confines the LLM to a single judgement step, with deterministic code on both sides of it.
3. A set of prompt constraints that are *verifiable* rather than aspirational (anonymisation, evidence-before-score, a materiality gate), achieving zero gate violations across 1,307 calls.
4. A three-layer validity framework—persistent neutralised random baselines, factor-adjusted alpha, and statistical-power expansion—and a record of six distinct measurement defects it caught.
5. A clean, well-powered negative result on the Item 1A diff-surprise approach, and a pre-registered test of an MD&A alternative.

1.5 Roadmap

Section 2 reviews the two source literatures briefly. Section 3 describes the data and how it changed across rounds. Section 4 specifies the full experimental design, including why each evaluation metric was chosen. Section 5 presents the eleven rounds in sequence. Section 6 synthesises where the project currently stands, and Section 7 states limitations and future work.

2 Related Work

2.1 Pre-LLM lineage

Four results motivate the design directly. Loughran and McDonald (2011) show that general-purpose sentiment dictionaries misclassify financial text and construct finance-specific word lists; these underpin the deterministic baselines used throughout this report. Campbell et al. (2014) establish that Item 1A risk-factor disclosure is firm-specific and informative rather than boilerplate, which is the reason to read Item 1A at all. Kravet and Muslu (2013) show that year-over-year changes in risk disclosure move volatility and trading activity. Cohen, Malloy and Nguyen (2020) show that *changes* in 10-K and 10-Q language predict returns, with the striking direction that firms whose filings change little outperform—the "Lazy Prices" effect, and the direct ancestor of the diff-based design here.

2.2 The LLM wave, organised by text source

The more informative way to read the recent LLM literature is to ask what file each study feeds the model (Table 1).

Table 1: Recent LLM equity-signal studies by text source.

Study	Text source	What is extracted
Lopez-Lira & Tang (2023/24)	Company news headlines	Good/bad/irrelevant score
Chen, Kelly & Xiu (2023/24)	Financial news articles	LLM embeddings → expected returns
Kim, Muhn & Nikolaev (2024)	Standardised numeric financial statements	Direction of future earnings
Kim, Muhn & Nikolaev (2023)	MD&A + earnings-call transcripts	Summaries; "bloat" measure
Huang, Wang & Yang (2023)	Reports, calls, analyst notes	Finance-domain language model
Chiu & Hung (2025)	**10-K MD&A**	Fine-tuned LLaMA-2 sentiment

The pattern is consistent: productive LLM signals concentrate on *timely* sources— news, transcripts, financial statements. Within the 10-K, MD&A has positive precedent (Chiu and Hung, 2025). Standalone Item 1A does not appear as a return signal in this literature. A methodological caution also recurs: Glasserman and Lin document look-ahead contamination, where a model's training data leaks knowledge of outcomes into ostensibly historical analysis.

2.3 Positioning

This study sits at the intersection of the "Item 1A + year-over-year change" line and the "LLM extraction" line. It differs from most of the LLM literature in three ways: it evaluates cross-sectionally rather than by event-study abnormal returns; it reports factor-adjusted alpha rather than raw or benchmark-relative returns; and it reports a negative result, which publication bias makes scarce.

3 Data

3.1 Universe

The universe is the 150 alphabetically-first current members of the S&P 500. This is a convenience sample and a known weakness (Section 7): membership is *current*, so the universe is survivorship-filtered by construction. Earlier rounds used a 42-name subset; the expansion to 150 is itself one of the experimental steps (Round 9–10).

3.2 Filings and section extraction

Filings are retrieved from SEC EDGAR. Two narrative sections are extracted:

- **Item 1A (Risk Factors)**, via a purpose-built parser. Coverage 99.3%.
- **Item 7 (MD&A)**, added in Round 11, via a three-stage extractor: Item 7 heading → in-report MD&A title (needed for filers such as JPM, GE, BAC) → Exhibit 13 fallback. Coverage 98.2% (1,946 by Item 7 heading, 99 by MD&A title, 63 via Exhibit 13, 38 not found).

Each current-year filing is paired with the same firm's prior-year filing to form a year-over-year pair. A whole-document word-shingle Jaccard similarity is stored for every pair as an *independent input-quality metric*—independent in the sense that it does not depend on the model's own output.

3.3 Prices, factors, and metadata

Monthly total returns are computed from daily price history; the prediction target is the one-month-forward return. Fama–French 3- and 5-factor returns plus momentum are used for factor attribution. Sector and market-capitalisation metadata are used for neutralisation. Both price history and metadata are *current*, not point-in-time—a limitation carried throughout.

3.4 How the data changed across rounds

The dataset is not constant across the study, and this matters for reading the results. Table 2 records the evolution.

Table 2: Evolution of the dataset across rounds. Changes in the data are themselves experimental steps, not incidental.

Rounds	Text source	Tickers	Filings	Months	Note
1–6	Item 1A	42	211	70	Pre-parser-repair text
7–8	Item 1A	42	211	70	Post-parser-repair (122/293 rows changed)
9	Item 1A	150	—	—	Feasibility dry run; 2 filing years only
10–10B	Item 1A	150	1,322	172	Deep universe; early years thin (see §5.10)
11	Item 1A + MD&A	150	2,146	185	Coverage defect repaired; 2011–2026

4 Experimental Design

4.1 Pipeline

The pipeline runs in five stages:

1. **Ingestion.** Download filings from EDGAR; extract Item 1A and MD&A; persist raw text with provenance (CIK, accession number, filing date).
2. **Deterministic diff.** Split each filing into sentence groups of roughly 700 characters. Greedily match current-year blocks to prior-year blocks by word-set Jaccard similarity. A block with no match above 0.50 is `added`; a match below 0.85 is `modified`; unmatched prior blocks are `removed`. Only changed blocks (up to 12,000 characters, longest first) proceed.
3. **LLM extraction.** The model receives *only* the changed text and scores it (Section 4.3).
4. **Signal assembly.** Deterministic code—not the model—converts the model's structured output into the final numeric signal.
5. **Evaluation.** Point-in-time alignment, portfolio construction, and the three validity checks.

The governing design principle is that **the LLM is confined to one step**. Deterministic code decides *what changed*; the model judges only *whether the change matters*; code produces the number.

Every responsibility handed to the model is reproducibility surrendered, so the model's uncertainty is deliberately isolated in a single, testable place.

4.2 Signal families

Deterministic lexicon signals. Transparent counts over finance-specific word lists, yielding `before_text_signal` and `enhanced_text_signal`, plus sector/size-residualised variants. These serve as the benchmark the LLM must beat.

Similarity signals. The Lazy-Prices construction: rank firms by whole-document year-over-year similarity, long high-similarity names.

LLM signals. Three generations: absolute severity scoring (Round 2), year-over-year relative scoring (Rounds 3–5), and diff-surprise scoring (Rounds 6–10B).

4.3 LLM configuration and prompt constraints

The model is DeepSeek (`deepseek-v4-flash`) at temperature 0 in enforced JSON mode, with schema validation and retry on malformed output. The prompt imposes five constraints, each chosen to be checkable after the fact:

1. **Evidence before score.** The model must enumerate up to five concrete changes—each with a scope tag (company-specific / sector-wide / market-wide), a direction, and an integer materiality 0–10—*before* emitting an aggregate. This suppresses reasoning-after-the-conclusion and makes every resulting position traceable to a cited change.
2. **Anonymisation.** No ticker, company name, or date appears anywhere in the prompt; sector is the only identity context. The model's training data spans the entire test period, so without this it is impossible to distinguish text comprehension from recall of outcomes.
3. **No numeric exemplars.** No example score values appear, to avoid anchoring.
4. **Explicit base rate.** The prompt states that risk sections grow through routine boilerplate maintenance, that growth in length is not a signal, and that for most filings the correct aggregate is zero.
5. **Materiality gate.** The aggregate must be zero unless at least one company-specific change has materiality ≥ 6. Violations are *recorded*, not silently corrected, so compliance is measurable.

The final signal is assembled in code:

$$s = \begin{cases} 0, & \text{if net_signal} = 0, \\ \text{net_signal} + 0.25\,(n_{\text{pos}} - n_{\text{neg}}), & \text{otherwise,} \end{cases}$$

where net_signal $\in [-5, 5]$ is the model's integer judgement and $n_{\text{pos}}, n_{\text{neg}}$ count company-specific changes by direction. Gated rows remain *exactly* zero; the correction that enforced this is itself a result (Section 5.5).

4.4 Point-in-time alignment

Each signal is stamped with the SEC `filing_date` and forward-filled to month-ends until superseded by the next filing. A signal is therefore never available before the document that generated it was public. The prediction target is the one-month-forward return.

4.5 Portfolio construction

The primary portfolio is a **sector-neutral rank-weighted long–short** book: within each sector, firms are ranked by signal, weights are assigned proportional to demeaned rank, and the book is dollar-neutral. Earlier rounds used top/bottom quintile tails; Rounds 1–4 broke ties by ticker order, which was replaced from Round 5 by fractional boundary handling so that tied scores split weight proportionally rather than being selected alphabetically.

4.6 Evaluation metrics, and why each is used

- **Rank information coefficient (rank IC).** The cross-sectional Spearman correlation between signal and forward return, averaged over months. This measures *ordering quality* independently of how the ordering is traded, which is essential because the same signal can produce very different Sharpe ratios under different weighting schemes (Section 5.6).
- **Sharpe ratio.** The headline risk-adjusted statistic for the long–short return stream. Reported alongside annualised return, maximum drawdown, and cumulative return so that a high Sharpe driven by a short benign period is visible.
- **Random-baseline percentile.** 500 Monte Carlo no-text signals are run through the identical pipeline on identical coverage. Crucially the random signals are *persistent*—held across months like a real annual-filing signal—and sector/size-neutralised. A baseline redrawn monthly would have different turnover from the real strategy, making the comparison invalid. This answers: *is this result distinguishable from luck?*
- **Factor-adjusted alpha.** The long–short stream is regressed on Fama–French 3- and 5-factor returns plus momentum. Because the book is dollar-neutral, the raw stream is regressed directly and the intercept is alpha. Standard errors are Newey–West HAC with 6 lags, because annual-filing signals hold positions for roughly twelve months and the resulting autocorrelation would otherwise understate standard errors and inflate t-statistics. This answers the strictly harder question: *is this alpha, or exposure to something already priced?*

The last two are not redundant. A pure market-beta portfolio is genuinely non-random and will score highly against random baselines while containing no alpha at all. The random baseline tests for luck; the factor regression tests for novelty. The gap between the two is exactly the factor-explained portion of the return.

4.7 Cost and reproducibility controls

Because each full extraction run has a monetary cost, three controls were adopted: a **pilot gate** (a 30-record run against pre-defined distributional pass criteria, with an automatic proceed / do-not-proceed decision) before any full run; a **mock mode** so pipeline development costs nothing; and **atomic checkpointing** (temporary file plus rename) with resumable extraction.

5 Results and Analysis

Each subsection below states the validity threat inherited from the previous round, the method used to close it, the result, and the threat that the result itself exposed.

5.1 Round 1 — Deterministic baseline and the evaluation scaffold

Objective. Establish transparent baselines and, more importantly, the machinery needed to judge any later result: a random no-text control and a neutralisation step.

Data. 42 tickers, 211 Item 1A filings, 70 monthly periods.

Table 3: Round 1: deterministic lexicon signals.

Signal	Ann. ret.	Sharpe	Max DD	Mean IC	Cum. ret.
`before_text_signal`	3.22%	0.31	−20.83%	2.48%	20.28%
`enhanced_text_signal`	3.10%	0.31	−13.92%	2.67%	19.50%
`enhanced_sector_size_neutral`	5.92%	**0.50**	−15.35%	2.00%	39.89%

Reading. The neutralised lexicon signal reaches the 88.8th percentile of the random distribution on Sharpe (empirical $p = 0.112$). It is suggestive and *not* significant at conventional levels. Reporting it as a success would have been the first available mistake.

Threat exposed → next step. Word counts cannot distinguish a new, material risk from routine template expansion. Round 2 tests whether semantic understanding helps.

5.2 Round 2 — Absolute LLM scoring: a task-framing failure

Method. The model scores each filing independently on severity, uncertainty, growth outlook, and margin pressure; a weighted combination forms the signal.

Table 4: Round 2: absolute LLM scoring versus baselines.

Signal	Ann. ret.	Sharpe	Max DD	Mean IC
`original_text_signal`	3.22%	0.31	−20.83%	2.48%
`deepseek_llm_text_signal`	−9.11%	**−0.51**	−53.63%	−2.15%
`random_no_text`	−0.34%	0.05	−34.71%	−0.81%

Reading. The LLM performs worse than the dictionary *and* worse than random. The diagnosis is not model capability but task definition. Item 1A is a section in which firms are legally obliged to enumerate risks; asking "how severe are the risks here" rewards firms that disclose incompletely and penalises those that disclose thoroughly. The model executed a logically defective instruction faithfully.

Threat exposed → next step. Absolute scoring is structurally confounded with disclosure completeness. Round 3 switches to within-firm year-over-year comparison, which differences that confound away.

5.3 Rounds 3–4 — Year-over-year scoring, and the sample-completeness check

Round 3 scored year-over-year change on 169 rows across 40 tickers and produced 5.56% annualised return at 0.48 Sharpe—apparently a substantial improvement. However, the prior-year context was incomplete for part of the universe.

Round 4 completed the universe to 211 rows and re-ran the identical method.

Table 5: Round 4: the same signal on the completed sample.

Signal	Ann. ret.	Sharpe	Max DD	Mean IC	Cum. ret.
`deepseek_yoy_relative_signal`	1.09%	0.15	−25.96%	−2.20%	6.55%
`enhanced_sector_size_neutral`	5.92%	0.50	−15.35%	2.00%	39.89%

Reading. Sharpe fell from 0.48 to 0.15 purely from completing the sample. The Round 3 result was an artifact of sample incompleteness, not a signal. This is the first instance of a pattern that recurs decisively in Round 10: *adding data removes an apparent effect.*

Threat exposed → next step. Before iterating further on the signal, the measurement apparatus itself needed auditing—if the ruler is biased, no comparison between rounds means anything.

5.4 Round 5 — Repairing the measurement, not the signal

Objective. This round deliberately changes *nothing* about any signal. It audits and repairs the evaluation code. Three defects were found, all of them directional rather than random noise:

1. **Tie-breaking by ticker order.** Tied scores were resolved alphabetically, systematically overweighting firms early in the alphabet. Replaced by fractional boundary handling.
2. **A non-persistent random baseline.** The control was redrawn each month, while the real signal—derived from annual filings—holds positions for about twelve months. A high-turnover control was being compared against a low-turnover strategy. Replaced by a persistent baseline.
3. **Unneutralised LLM output.** Raw LLM signals carried sector and size exposure, so part of what was being measured was an industry bet. Residualised.

Table 6: Round 5: all prior results recomputed under corrected measurement.

Signal	Ann. ret.	Sharpe	Max DD	Cum. ret.
`enhanced_sector_size_neutral`	5.32%	**0.48**	−16.55%	35.30%
`deepseek_absolute_sector_size_neutral`	−3.53%	−0.20	−41.53%	−18.94%
`deepseek_yoy_relative_sector_size_neutral`	0.51%	0.10	−28.21%	3.02%
`random_no_text_persistent_neutralized`	−4.97%	−0.29	−39.35%	−25.72%

Reading. The round produced no new signal and no new result—and it is arguably the most important round in the study. Every comparison made before it had been measured with a biased instrument. The year-over-year LLM signal, once its sector tilt was removed, settles at 0.10 Sharpe: essentially nothing.

Threat exposed → next step. With a trustworthy ruler, attention returns to the model. Inspection of the raw outputs showed scores clustering tightly on the example values printed in the prompt.

5.5 Round 6 — Prompt redesign under a pilot gate

Stage A: remove numeric anchors. Under the old prompt, 60% of `yoy_direction_score` values sat at exactly −1.00, and roughly a quarter of other component scores sat on their respective printed exemplars. Stage A removed all numeric examples and ran a 30-record pilot.

The anchors vanished—exemplar shares fell to zero—but the signal's distinct-value count collapsed from 22 to 8: the model had merely retreated to a coarser $\{-1, 0, 1\}$ rubric. The pilot gate returned `do_not_full_run_yet` and **the full run was never paid for**. The gate also diagnosed the problem: the issue was not wording but task design.

Stage B: change the task. The redesign feeds the model only the deterministic diff, asks for company-specific *investable surprise* rather than risk severity, requires evidence before score, and imposes the materiality gate. The pilot passed (21/30 zeros, zero gate violations). On the full 211 rows: 64% zeros, 75 non-zero rows of which 56 negative and 19 positive, and zero gate violations.

Table 7: Round 6 Stage B: diff-surprise signal.

Signal	Ann. ret.	Sharpe	Max DD	Cum. ret.
`deepseek_diff_surprise_signal`	−1.91%	−0.14	−13.76%	−10.64%
`deepseek_diff_surprise_sector_size_neutral`	4.46%	**0.43**	−21.73%	28.98%
`enhanced_sector_size_neutral`	5.32%	0.48	−16.55%	35.30%

Post-run code review: a defect worth reporting. The prompt required a zero aggregate absent a high-materiality company-specific change, and the model complied perfectly—zero violations. But the downstream assembly formula applied the intensity term $0.25(n_{\text{pos}} - n_{\text{neg}})$ *unconditionally*. On 49 of 211 records the model reported "no signal" and the surrounding code supplied a direction anyway. Enforcing the gate in code improved the result from 0.33 to 0.43 Sharpe (21.79% to 28.98% cumulative): the sub-threshold direction had been adding noise, not information. The general lesson—*a rule written in a prompt is a request; a rule enforced in code is a control*—was applied to all subsequent constraints.

Threat exposed → next step. The diff-quality instrumentation showed 12 pairs with similarity below 0.20. Manual inspection showed these were not genuine rewrites.

5.6 Round 7 — Parser repair and portfolio-construction sensitivity

The defect. The Item 1A parser treated mid-sentence cross-references such as "see Item 1A. Risk Factors of this Annual Report" as section starts. Lacking a nearby terminating heading, the extractor then returned MD&A, properties, or litigation text instead of the risk section. **122 of 293 records—42%—had been fed the wrong text**, and the model had returned well-formed, plausibly-reasoned JSON for every one of them. Nothing errored; nothing flagged.

All Round 6 LLM results were voided and the extraction re-run on repaired text.

Table 8: Round 7: post-repair results. Note the benchmark's sign reversal.

Signal	Ann. ret.	Sharpe	Mean rank IC	Cum. ret.
`enhanced_text_signal`	4.76%	0.32	0.10%	31.18%
`enhanced_sector_size_neutral`	−4.55%	**−0.33**	−1.45%	−23.81%
`deepseek_diff_surprise_sector_size_neutral`	0.15%	0.07	2.88%	0.88%

Portfolio-construction sensitivity. Holding the signal fixed, four weighting schemes were compared. All four share an identical mean rank IC of 2.88%.

Table 9: Round 7: identical signal, four constructions. Rank IC is 2.88% in every row.

Construction	Ann. ret.	Sharpe	vs-random pct.
Top/bottom 20% tails	0.15%	0.07	57.0
Top/bottom 30% tails	2.61%	0.28	74.4
Rank-weighted	3.50%	0.39	80.8
Sector-neutral rank-weighted	5.69%	**0.57**	91.8

Reading. Two things stand out. First, the deterministic benchmark *flipped sign* on repaired text—a signal that reverses when an input bug is fixed is not stable enough to trust on this sample. Second, Sharpe moved by a factor of eight across constructions with no change whatsoever in signal quality: a substantial part of what had been attributed to the model was downstream weighting. Sector-neutral rank weighting was adopted as the primary construction from here on—and it must be noted that this choice was made *after* seeing these results, which weakens any significance claim attached to it.

Threat exposed → next step. A 91st-percentile result against random baselines still cannot distinguish alpha from priced factor exposure.

5.7 Round 8 — Factor attribution

Method. The Round 7 long–short stream is regressed on FF3 and FF5 plus momentum, with Newey–West HAC standard errors at 6 lags.

Result. FF3+MOM annualised alpha of 6.65%, $t = 2.05$, $p = 0.045$.

Reading. The "91st percentile versus random" narrows to "marginally significant versus factors". The gap between the two numbers is precisely the factor-explained portion of the return. Taken at face value this clears a 95% bar—but three caveats sit on top of it: the sample is 42 names and 70 months, the portfolio construction was selected post-hoc, and the deterministic control on the same data is unstable. When the control is unstable, the p-value is describing noise precisely.

Threat exposed → next step. Statistical power, not prompt quality, is the binding constraint. Further prompt iteration on 42 names would be tuning inside noise.

5.8 Round 9 — Universe-expansion feasibility

Objective. Before spending on a large extraction, verify that the pipeline scales. This round makes *no API calls*.

Result. The parser extends cleanly: at 75 tickers, median pair similarity is 0.794 with no pairs below 0.30; at 150 tickers, median similarity is 0.78 with 12 pairs below 0.30. A 150-pair stratified LLM pilot showed a healthy output distribution. However, the pilot window covered only two filing years—wide but shallow—so no backtest was possible from it.

Reading. Feasibility confirmed, and the expensive step deferred until the cheap step had de-risked it.

5.9 Rounds 10 and 10B — The powered test

Objective. Run both the deterministic benchmark and the full LLM signal through identical construction on a sixfold larger sample: 150 tickers, 1,322 filings, 1,307 year-over-year pairs, 172 monthly periods, 500 random trials. Round 10 (deterministic) requires no API spend and was run first.

Table 10: Rounds 10/10B: sector-neutral rank-weighted, identical construction, 172 monthly periods.

Signal	Rank IC	Sharpe	Cum.	FF3+MOM alpha	vs-rand.
`deepseek_diff_surprise` (LLM)	−0.040	**−0.83**	−73.4%	−8.94%, $t = -2.54$, $p = 0.012$	0.0
`enhanced_text_signal` (det.)	+0.033	**+0.68**	+136.5%	+5.11%, $t = 1.86$, $p = 0.065$	99.0
`before_text_signal` (det.)	+0.026	+0.62	+123.8%	+4.01%, $t = 1.57$, $p = 0.119$	99.2

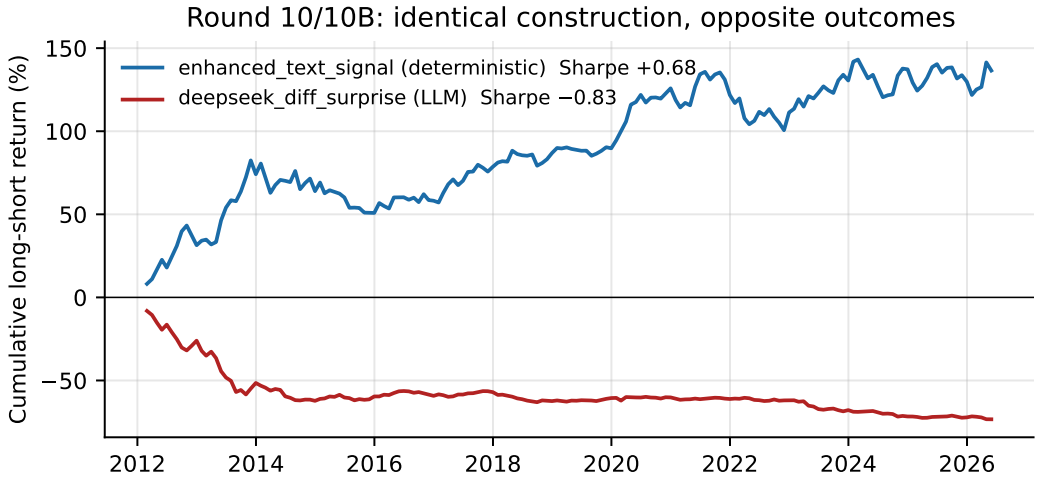


Figure 1: Round 10/10B cumulative long–short returns. The two signals pass through the same parser, the same point-in-time alignment, the same sector-neutral rank-weighted construction, and the same factor model.

Reading. This is the decisive result of the study, and it cuts in two directions at once.

The LLM signal is not merely uninformative; it is *anti-predictive*. Negative rank IC, significantly negative factor alpha, and a 0th percentile against 500 random baselines—it is worse than every random signal tested. Simultaneously, the deterministic dictionary signal, which had looked unstable and even negative at 42 names, emerges as genuinely positive. **One change—sample size—revealed two truths that the small sample had been hiding at once.**

The Round 7 LLM advantage (+0.57 Sharpe, alpha $p = 0.045$) is therefore an artifact of small sample plus post-hoc construction selection. This is the point at which the answer stopped changing as the test got stricter.

Mechanism. The output distribution identifies the cause (Figure 2). Of 1,307 calls, 638 produced a non-zero score and **75.7% of those were negative**. More tellingly, the distribution is *asymmetric in range*: the model used the full negative half of its permitted scale down to −5, but never exceeded +3 on the positive side. It equated "this firm disclosed bad news" with "this firm will underperform"—a correct reading with an incorrect causal direction, since disclosure is often already priced and resolution of uncertainty can itself be favourable. Critically, this is invisible in any single output: every individual response was well-reasoned and defensible. It appears only in aggregate.

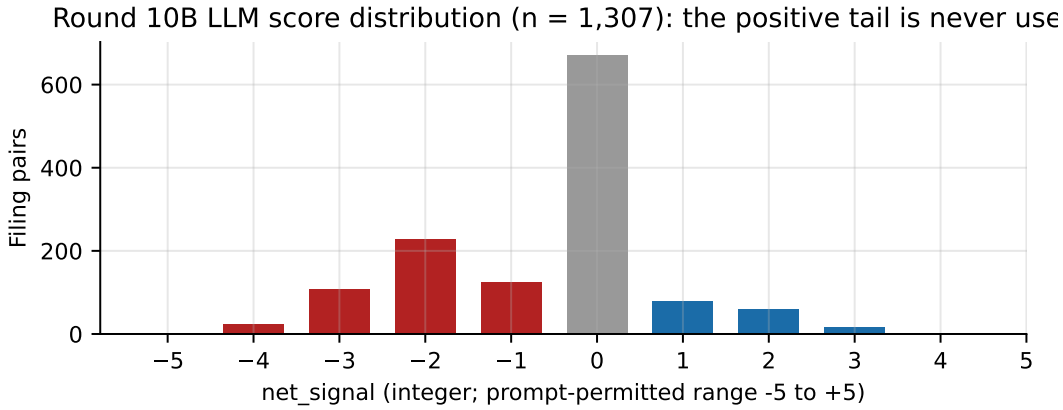


Figure 2: Round 10B score distribution. The prompt permitted −5 to +5; the model used the entire negative range and never the top of the positive range.

Infrastructure note. During this extraction a long-running job was terminated mid-write, truncating the output file and losing roughly 700 already-paid extractions irrecoverably. The checkpoint had been rewriting the entire file, including a full-text column, on every row—so the write itself was the failure point. Checkpointing was moved to atomic write (temporary file plus rename) and the full-text column dropped, shrinking the file roughly a hundredfold. All 1,307 extractions were re-run.

Threat exposed → next step. The conclusion is specific to Item 1A. The literature suggests the productive sources are timelier ones—and within the same 10-K, MD&A has positive precedent.

5.10 Round 11 — Changing the text source, and a coverage defect

Objective. Test whether the source, rather than the method, is the binding constraint, using the lowest-cost available source change: MD&A from the same filings. No API calls; deterministic signals only. Two hypotheses were **fixed before results were computed**:

- **H1** `mdna_similarity`: the Lazy-Prices direction—high year-over-year similarity goes long.
- **H2** `mdna_enhanced`: the lexicon signal applied to MD&A.

A coverage defect found while preparing this round. The earlier downloader read only the `recent` array of the SEC submissions API, which holds roughly the last 1,000 filings of *all* types per

company. For large filers this silently dropped early 10-Ks. The Round 10 "2012–2026" universe was therefore not uniformly deep: the cross-section contained as few as four names in 2012 and did not reach 100 until after 2019 (Figure 3). Re-downloading with `include_older=True` yielded 121–150 tickers per year for 2011–2026 and raised the corpus from 1,322 to 2,146 10-Ks. A related trap: the SEC ticker map now resolves XOM to a 2026 holding company with no 10-K history, so Round 11 keys on CIK from the universe file instead of ticker.

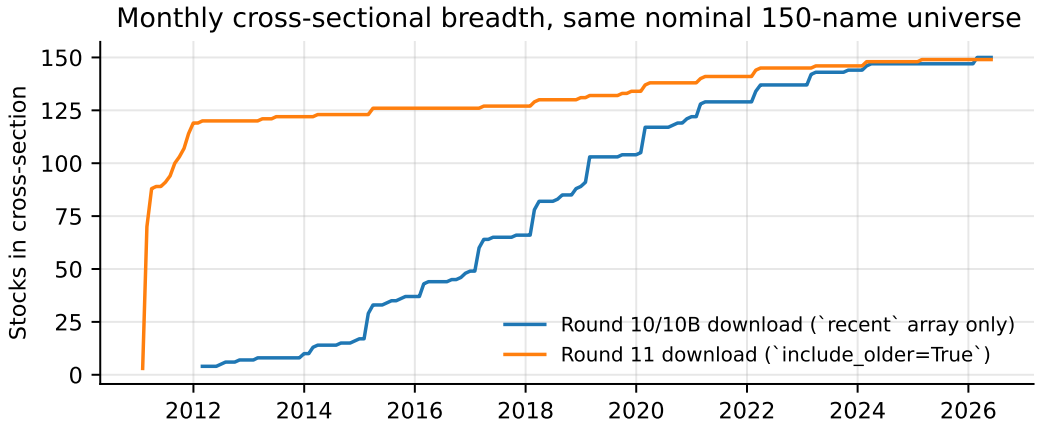


Figure 3: Monthly cross-sectional breadth under the two download methods, for the same nominal 150-name universe. The Round 10/10B sample was a ramp, not a panel.

Table 11: Round 11: MD&A and Item 1A signals on the full-coverage universe, sector-neutral rank-weighted.

Signal	Role	Sharpe	vs-rand.	FF3+MOM alpha	Verdict
`mdna_similarity`	H1	−0.72	0.4	−3.6%, $t = -2.95$	Significant, **sign reversed**
`mdna_enhanced`	H2	+0.20	76.8	+0.3%, $t = 0.14$	Fail
`item1a_enhanced`	exploratory	+0.49	95.4	+2.4%, $t = 1.88$	Survives, weaker than Round 10
`item1a_similarity`	exploratory	+0.25	81.2	+0.3%, $t = 0.29$	2019–2026 only
`mdna_tone_change`	exploratory	+0.00	52.4	≈ 0	Fail

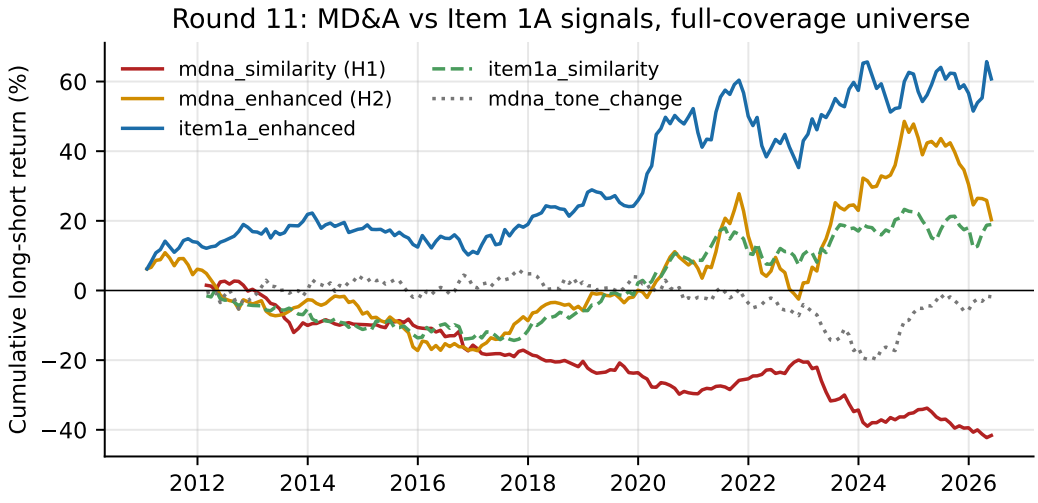


Figure 4: Round 11 cumulative long–short returns across the five signals.

Reading.

H2 fails outright. A dictionary on MD&A produces 0.20 Sharpe and essentially zero alpha. MD&A tone change likewise produces nothing. The deterministic case for MD&A as a source is not made.

H1 is significant with the wrong sign. High-similarity firms *under*perform, the opposite of Lazy Prices, at -0.72 Sharpe and alpha $t = -2.95$. The sign is robust within sample: restricting to pairs extracted by the same method in both years gives -2.83; restricting to the Item 7 layout gives -2.70; residualising similarity on absolute log length change gives -2.19; both sub-periods are negative. The most plausible explanation is nonetheless **survivorship bias**: in a universe of *current* index members, firms whose MD&A changed substantially (acquisitions, new segments, rapid growth) and which remain in the index today are survivors by construction. "Changers outperform" may be an artifact of who is in the sample. Since the sign was also not the pre-registered direction, this is a post-hoc finding pending an out-of-sample test on a point-in-time universe.

The Item 1A benchmark weakens but survives. On full coverage it falls from 0.68 to 0.49 Sharpe, remaining at the 95th percentile versus random with FF5+MOM alpha $t = 2.12$. The earlier, higher figure was partly a product of the thin early cross-section.

A measurement caveat discovered in this round, applying retroactively. The random rank-IC percentile statistic is optimistic. Random signals are sector/size-neutralised before their IC is computed, whereas the target signal is not, so the random IC spread (sd ≈ 0.006) is roughly half the target's own time-series standard error (≈ 0.013). IC percentiles therefore overstate significance. Sharpe percentiles are unaffected, because the portfolio is sector-neutral on both sides. All IC percentiles reported in earlier rounds should be read with this in mind; Sharpe percentiles and t-statistics are the reliable statistics.

6 Synthesis: Where the Project Currently Stands

6.1 The trajectory

Figure 5 plots the headline Sharpe of the best LLM signal and the deterministic benchmark by round. The universe and construction change between rounds, so the levels are not strictly comparable—but that is the point. The movement tracks changes in *measurement*, not changes in the underlying phenomenon.

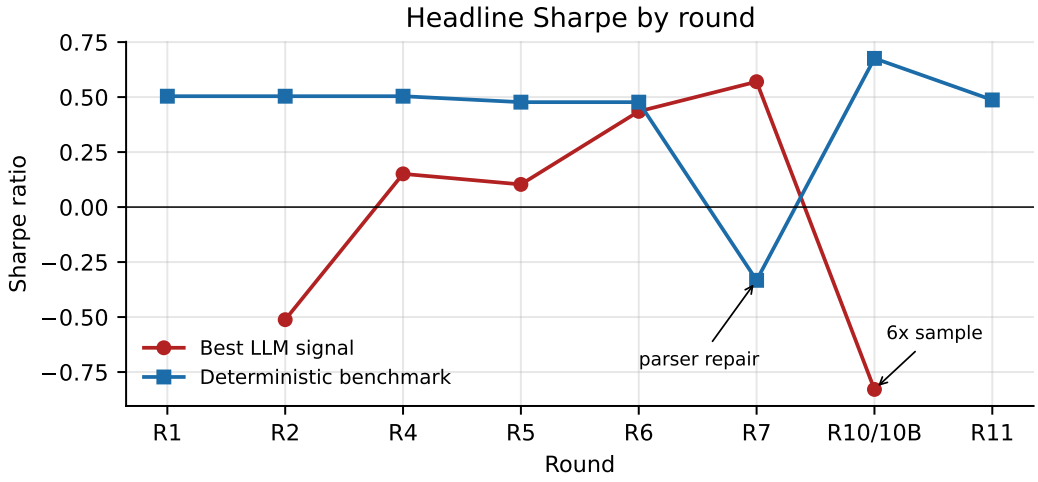


Figure 5: Headline Sharpe by round. Universe size and portfolio construction change between rounds; the two largest moves coincide with the parser repair (Round 7) and the sixfold sample expansion (Round 10B).

6.2 Validity threats closed

Table 12: Each round's validity threat, the method used, and the effect on the conclusion.

Round	Threat	Method	Effect
1	No control for luck	Random no-text baseline	Results gain a reference distribution
2	Word counts lack semantics	LLM absolute scoring	Identifies a task-framing defect
3–4	Incomplete sample	Complete the universe	0.48 → 0.15 Sharpe
5	Biased measurement instrument	Tie handling, persistent baseline, neutralisation	All prior rounds recomputed
6A	Prompt anchoring	30-record pilot gate	Full run avoided; task identified as the issue
6B	Task shape	Diff + investable surprise	Healthy distribution, zero gate violations
6 rev.	Code overrode model abstention	Enforce gate in code	0.33 → 0.43 Sharpe
7	42% of inputs were wrong text	Parser repair, full re-run	Round 6 results voided
7b	Construction absorbing the signal	Four-way variant test	0.07 → 0.57 at constant IC
8	Factor exposure read as alpha	FF3/FF5+MOM, Newey–West	91st pct. → marginal alpha
9	Unknown scalability	Zero-API dry run	Spend de-risked
10/10B	Insufficient power	6× sample	Two hidden truths revealed
11	Source-specific conclusion; coverage defect	MD&A + `include_older`	1,322 → 2,146 filings

6.3 What holds and what does not

Falsified. The LLM diff-surprise approach on Item 1A, as designed. At full power it is significantly negative, worse than every random baseline, and beaten by a word-count dictionary on identical data. The mechanism—structural negativity bias arising from per-item absolute scoring—is identified and is not a wording problem.

Not supported. MD&A as a deterministic source. Neither the lexicon nor tone change produces a signal, so the bar for an LLM step on MD&A is correspondingly high.

Holds, weakly. The Item 1A lexicon signal: 0.49 Sharpe on full coverage, 95th percentile versus random, FF5+MOM alpha $t = 2.12$. It carries positive market beta, so part of the return is exposure rather than alpha, and it is only marginally significant.

Unresolved and interesting. The reversed MD&A similarity effect, which is statistically strong and robust to several within-sample checks but is confounded with survivorship by construction.

6.4 Transferable findings

1. **Most LLM failures here were context or plumbing failures, not prompt failures.** The parser defect, the coverage defect, and the code-override defect together account for more swings in the headline number than every prompt revision combined.
2. **A model cannot report that its input was bad.** Independent, per-record input-quality instrumentation is not optional.

3. **With no ground truth, the output distribution is the quality check.** The negativity bias was invisible per-response and obvious in aggregate.
4. **A rule in a prompt is a request; a rule in code is a control.** The model complied perfectly with a constraint the surrounding code then violated.
5. **Evaluate the model output and the consumption layer separately.** An eightfold Sharpe swing at constant rank IC is a property of the weighting scheme, not the model.
6. **Statistical power is a validity threat, not a nice-to-have.** The same expansion that falsified the LLM signal validated the benchmark.

7 Limitations and Future Work

7.1 Limitations

1. **Survivorship and selection bias.** The universe is the alphabetically-first 150 *current* S&P 500 members. This is the most serious threat to *interpreting any positive result*, and it bears directly on the Round 11 H1 finding. Because the bias inflates rather than suppresses apparent performance, the negative results reported here are not weakened by it.
2. **Metadata is not point-in-time.** Sector and market-capitalisation data are current; price history is for currently-listed tickers.
3. **Post-hoc construction choice.** Sector-neutral rank weighting was adopted after observing Round 7 variant results, weakening significance claims attached to it.
4. **Gross results only.** No transaction costs, shorting costs, or capacity constraints.
5. **IC percentiles overstate significance** for the reason given in Section 5.10; Sharpe percentiles and t-statistics should be preferred.
6. **Single model family.** All LLM results use one provider. Whether the negativity bias generalises across model families is untested.
7. **Retroactive re-runs.** Rounds 1–10B have not been re-run on the full-coverage Round 11 corpus, so their levels are not directly comparable to Round 11's.

7.2 Future work, in priority order

The ordering below follows the staging discipline that Round 9–10 established: run the cheap, high-information experiment first, and spend on rigour once there is something worth making rigorous.

1. **Change the text source.** This is first because both narrative sources available inside the 10-K have now been tested and neither carries a usable signal: the LLM on Item 1A is significantly negative at full power, and on MD&A neither the lexicon nor tone change produces anything. The recent literature concentrates almost entirely on *timelier* sources—earnings-call transcripts, news, and standardised financial statements—and this study has never tested an LLM against one of them. Everything else on this list optimises *within* a source family that has so far shown no signal.

A new source also screens cheaply on the existing universe, and the asymmetry of survivorship bias makes that screen informative rather than merely convenient. Survivorship inflates apparent performance, so a *null* on the current universe is strong evidence of no effect, while a *positive* must be held back until item 2 is done. The screen is therefore safe in the direction that matters.

Within this item, two design choices follow directly from the failures documented above, and should be adopted rather than re-derived:

- **Listwise, not per-item.** Per-item absolute or incremental scoring is structurally prone to the negativity bias of Section 5. Asking the model to *order* firms within a sector-year cannot inherit a uniform negative tilt by construction.
- **Extractor, not judge.** Where feasible, use embeddings with a controlled statistical layer (the Chen–Kelly–Xiu pattern) rather than having the model emit a score. This is far easier to regularise, to validate, and to govern.

2. **Point-in-time universe, including delisted firms.** This is the gate that must be passed before *any* positive result is believed—including the Round 11 H1 reversal, which is statistically strong but confounded with survivorship by construction. It is second rather than first only because there is currently no live positive result that depends on it: resolving H1 alone would mean spending the project's next large effort confirming a post-hoc finding that the analysis already attributes to bias. It becomes first the moment item 1 produces something positive, or the moment the H1 effect is to be claimed rather than noted.

3. **Repair the IC baseline.** Apply identical neutralisation to the target signal before computing IC percentiles, and recompute the affected statistics. This is cheap and independent of everything else, so it can be done at any point.

4. **Beta-neutral construction of the surviving signal.** Build a strict dollar- and beta-neutral version of the Item 1A lexicon signal to isolate how much factor-independent alpha remains once market exposure is removed. This is the most defensible publishable claim available from the current data.

5. **Re-run earlier rounds on the full-coverage corpus.** Rounds 1–10B were computed before the filing-coverage defect of Section 5.10 was found, so their levels are not directly comparable to Round 11's. This is bookkeeping rather than discovery, but it is required before the cross-round trajectory is presented as evidence of anything.

7.3 What is explicitly not worth doing

Further prompt iteration on Item 1A diff-surprise. The failure mechanism is structural, six alternative explanations have been closed, and the result is significantly negative at full power. Additional spend on that path would buy precision about a conclusion that is already established.

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A Reproduction

All results are regenerated from the repository without manual steps.

Table 13: Entry points by round.

Round	Command
7	`python3 scripts/run_round7_item1a_repair_comparison.py`
7b	`python3 scripts/analyze_round7_error_and_portfolio_variants.py`
8	`python3 scripts/run_round8_factor_regression.py`
9	`python3 scripts/run_round9_universe_dry_run.py`
9B	`python3 scripts/run_round9b_llm_pilot.py`
10	`python3 scripts/run_round10_deep_backtest.py`
10B	`python3 scripts/run_round10b_llm_backtest.py`
11	`python3 scripts/download_round11_mdna.py` then `run_round11_mdna_backtest.py`
Figures	`python3 report/make_figures.py`

Rounds 10 and 11 require no API credentials; Round 10B re-runs 1,307 paid LLM extractions.